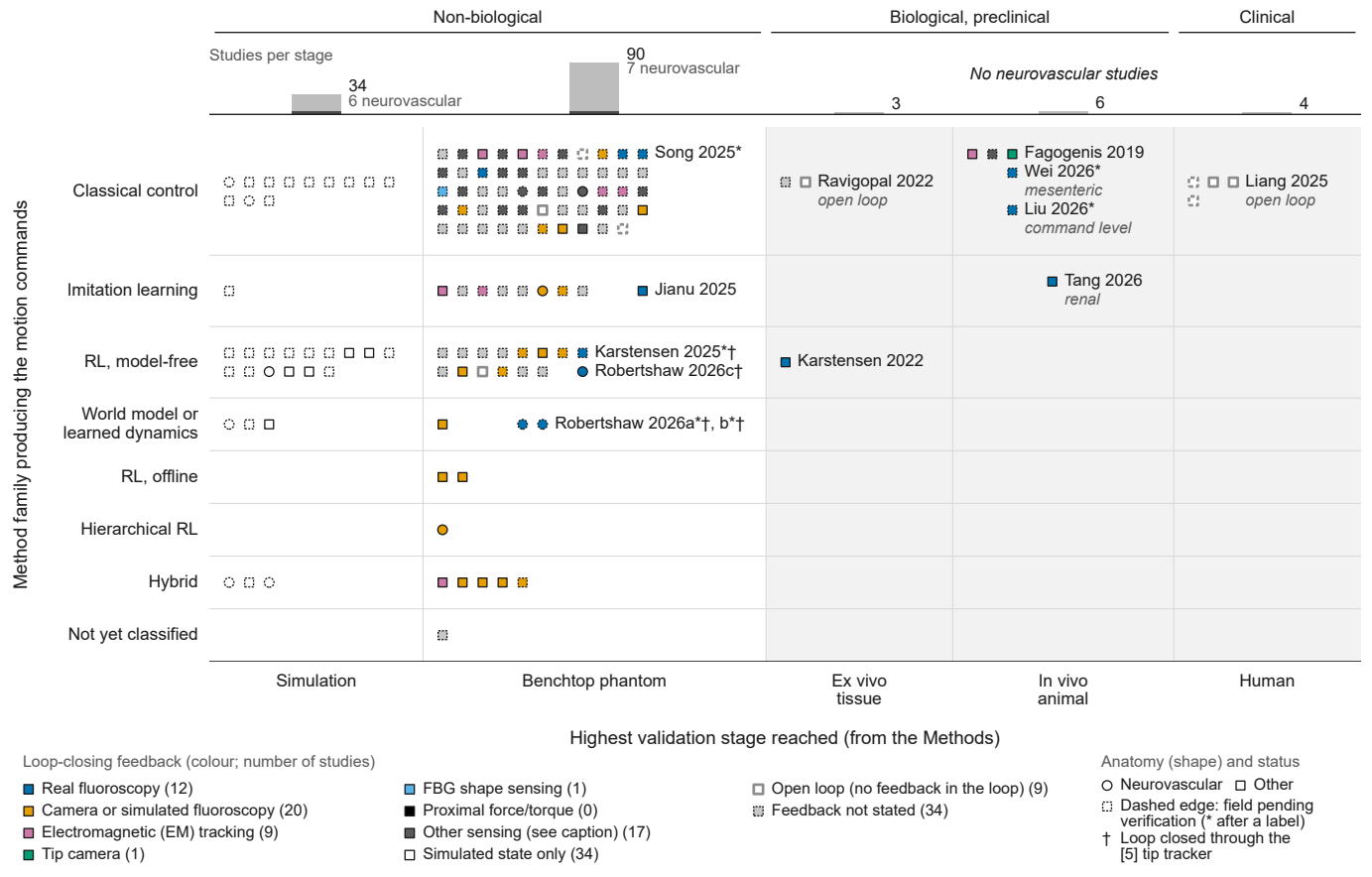
Non-biological
Biological, preclinical
Clinical
Studies per stage
34
6 neurovascular
90
7 neurovascular
No neurovascular studies
3
6
4
Method family producing the motion commands
Classical control
Imitation learning
RL, model-free
World model or learned dynamics
RL, offline
Hierarchical RL
Hybrid
Not yet classified
Song 2025*
Ravigopal 2022
open loop
Fagogenis 2019
Wei 2026*
mesenteric
Liu 2026*
command level
Liang 2025
open loop
Jianu 2025
Tang 2026
renal
Karstensen 2025*†
Robertshaw 2026c†
Karstensen 2022
Robertshaw 2026a*†, b*†
Simulation
Benchtop phantom
Ex vivo tissue
In vivo animal
Human
Highest validation stage reached (from the Methods)
Loop-closing feedback (colour; number of studies)
Real fluoroscopy (12)
Camera or simulated fluoroscopy (20)
Electromagnetic (EM) tracking (9)
Tip camera (1)
FBG shape sensing (1)
Proximal force/torque (0)
Other sensing (see caption) (17)
Simulated state only (34)
Open loop (no feedback in the loop) (9)
Feedback not stated (34)
Anatomy (shape) and status
Neurovascular
Other
Dashed edge: field pending verification (* after a label)
† Loop closed through the [5] tip tracker